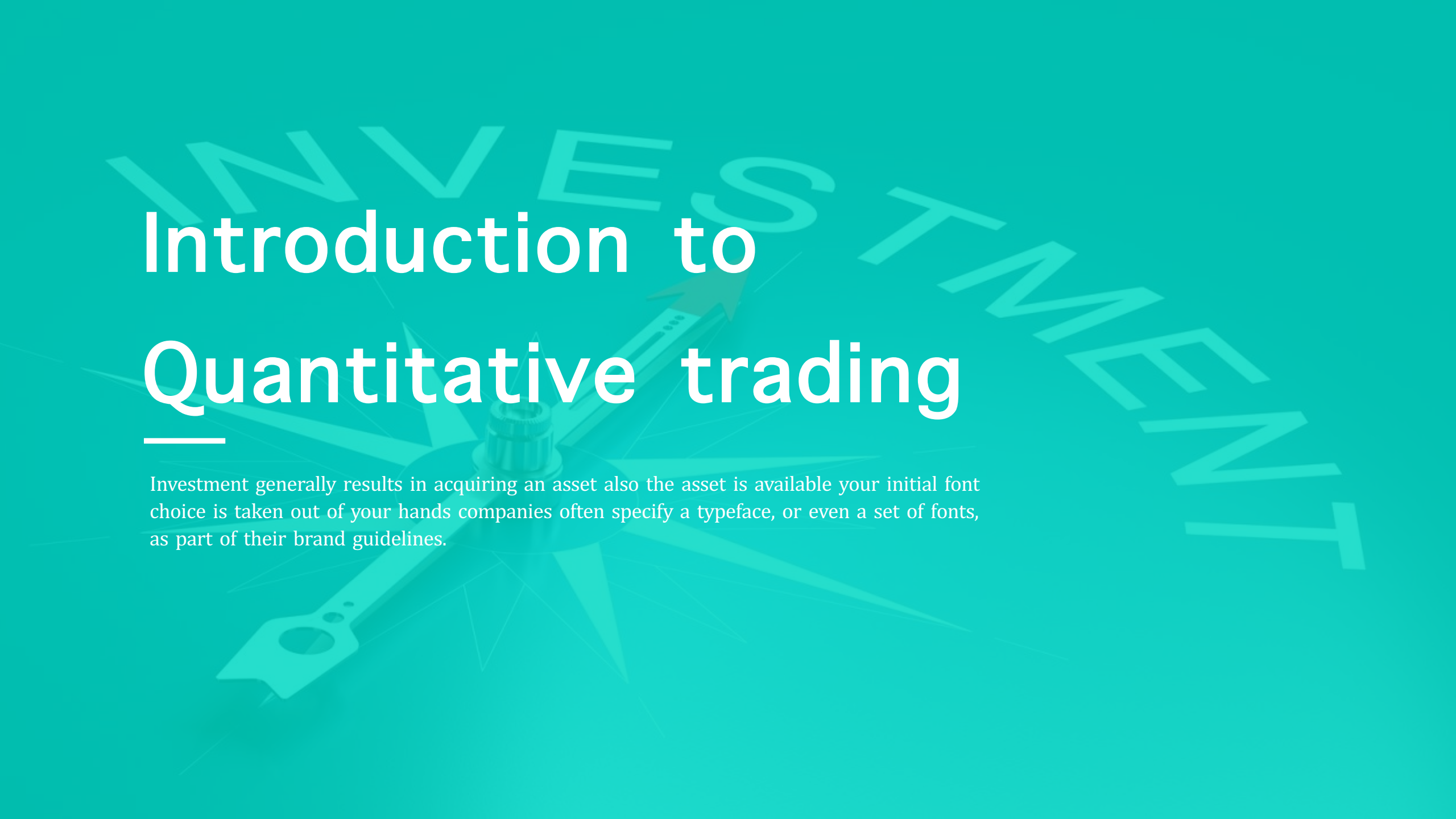


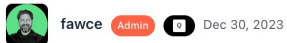


Introduction to Quantitative trading

Investment generally results in acquiring an asset also the asset is available your initial font choice is taken out of your hands companies often specify a typeface, or even a set of fonts, as part of their brand guidelines.

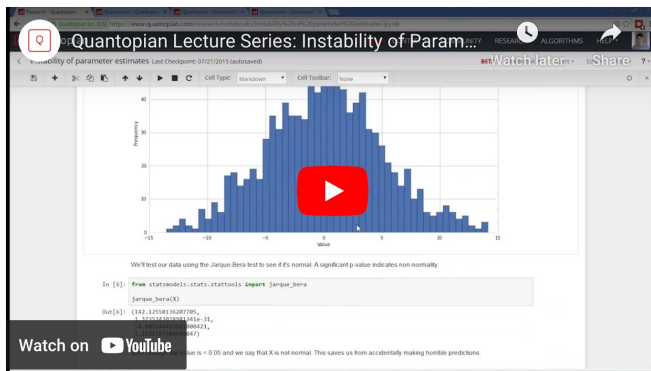
Quantopian Lectures are Back!

Share



55 notebook based lessons have been curated into an archive with summaries and links to the original notebooks, ports to other platforms, and embedded videos. The full series is available in the sidebar under the Quantopian Lectures heading.

Sample Lecture: Instability of Estimates



This Jupyter notebook delves into the instability of parameter estimates in financial models. It explains that parameters, such as mean and variance, are crucial for

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An In-depth Exploration of Systematic Trading Principles

3.7 ★★★★★ (28 คะแนน) ผู้เรียน 366 คน

สร้างโดย [Hudson and Thames Quantitative Research](#)

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สำหรับผู้ที่สนใจ

“Quant trading was
dominated by arbitrage
strategies that”

คุณ Jack



Trading VS Investing

Trading VS Investing

Trader VS Portfolio Manager

Alpha VS Outperformance

Alpha

Trading VS Investing

Their goal is to generate a positive return that is independent of overall moves in the market. This hedge return is called Alpha.

The term Alpha is also widely used by portfolio managers to refer to their outperformance or return above a benchmark.

Trading VS Investing

Frequent VS Portfolio

Transaction rebalancing

Short term VS long term

Perfect Trading Strategy?

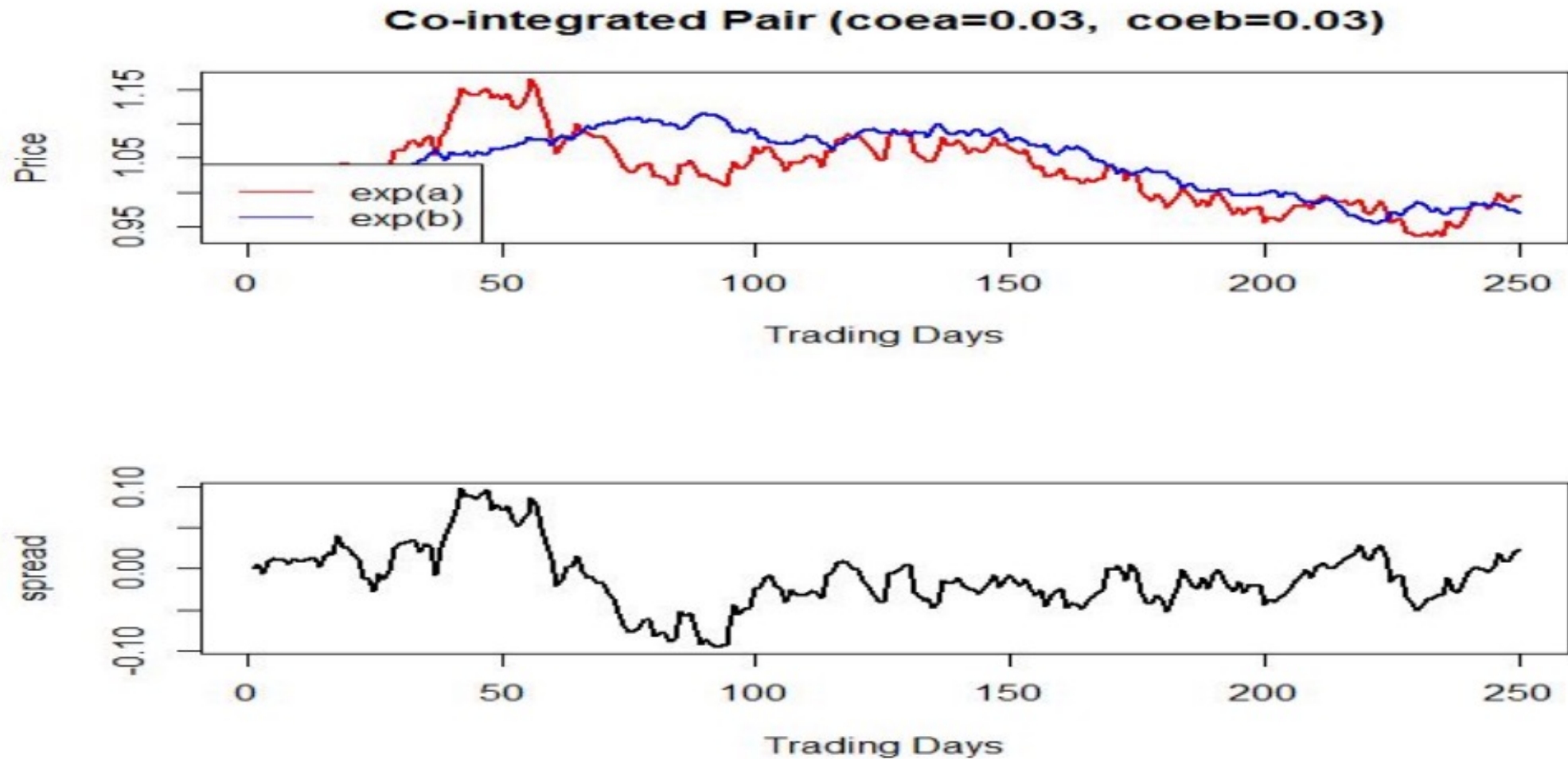
Quant Strategies

- Create a set of rules to generate trade orders and manage position risk with minimal intervention
- Identify statistically significant and repeatable market behavior and exploit it to generate profits.
- Can be low-frequency (weekly, daily) to high-frequency (seconds, milliseconds,...)

Quant Strategies

- Forecasting methods attempt to predict the future value of a financial instrument* or direction of a spread
- Mean Reversion trades on the deviation of a price spread between two or more financial instruments.
- Utilizes correlation and cointegration tests to identify significant mean reverting behavior.

Quant Strategies: Correlation vs Cointegration



Momentum or “trend following” depends on slow diffusion of information vs instantaneous diffusion assumed by the Efficient Market Hypothesis

Quant Strategies

1. Markets tend to underreact to news about a company
2. Markets tend to overreact
3. Momentum stock tend to be riskier
4. The outperformance of momentum stocks has drawn investors which has pushed up prices.

Quant Strategies

High Frequency Trading exploits millisecond and sub-millisecond market microstructure inefficiencies.

- Focuses on infrastructure flaws (uneven playing field) rather than longer term predictions of asset prices
- Models behavior patterns of major players in a market
- Seeks to uncover large orders hidden by stealth execution strategies

Quant Strategies

Examples of High Frequency Trading Strategies

- An investment fund wants to purchase a large block of stock.
- Uses stealth execution strategy to mask the order but not perfectly.
- Other market players detect the strategy and “jump ahead” of the order with their own buy orders
- Spoofing “fake” orders

Quant Trading Advantages

- Takes the emotion out
- Unambiguous path/strategy
- Mathematically optimal

Quant Trading Challenges

Trading is Risky

- Firms 'blow up' all the time
- Knight Capital lost ~\$478M in less than one hour of trading and had to be sold to another trading firm